

Curriculum vitae

PERSONAL DETAILS

Full Name: Md Abdullah-Al Mamun

Researcher ID:  [ORCID](#)  [Google Scholar](#)  [Sciprofiles](#)  [ResearchGate](#)

Online Profiles:  [LinkedIn](#) |  [GitHub](#) | [Portfolio](#)

Contact Details:

Location: Salo, Finland

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Date of CV: September 2026

DEGREES

Master of Science (M.Sc.) in Computing Sciences (Specialisation in Data Science)

Tampere University, Finland | **Completion Date:** 4 May 2024

Master's Thesis:

Map out the best BI tool and SPC software for a case company: selection and rationalization process.

Supervisors: Professor Martti Juhola (Tampere University), Timo Poranen (Lecturer, Tampere University), Henna Strömberg (Valmet Automotive EV Power Oy, Salo, Finland) | Published in [Trepo](#).

Relevant Coursework:

Pattern Recognition and Machine Learning (5/5), Data Mining (5/5), Machine Learning Algorithms (5/5), Dimensionality Reduction and Visualization (4/5), Statistical Inference (4/5).

Bachelor of Science (B.Sc.) in Biotechnology and Genetic Engineering

Mawlana Bhashani University of Science and Technology, Bangladesh | **Completion Date:** 30 January 2010

Core Coursework:

*Molecular Biology, **Bioinformatics**, Biostatistics, Biochemistry, Genetics.*

Bachelor of Science (B.Sc.) in Information Technology (Software Engineering Major)

Metropolia University of Applied Sciences, Finland | **Completion Date:** 18 December 2017

Relevant Coursework:

Advanced Programming (Python, Java), Software Engineering, Web Development, Android Application Development.

TECHNICAL SKILLS

Python, R, MATLAB, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, SQL, Git, Power BI, Excel.

LANGUAGE SKILLS

● **English:** Professional and Fluent ● **Finnish:** A2.2 (CEFR self-assessment) ● **Bengali:** Native

WORK EXPERIENCE

Early Career Researcher (ECR)

May 2024 – Present

- Published **10 peer-reviewed articles** (9 in Q1 journals) focusing on machine learning applications for groundwater quality and flood resilience.
- Led technical implementation using Python, R, and PyTorch for geospatial geostatistics and spatial hot-spot analysis.
- Contributed to the scientific community as a peer reviewer for the journal Environmental Impact Assessment Review (Elsevier).

Peer Reviewer

April 2026 – May 2026

[Environmental Impact Assessment Review](#) (Elsevier)

- Reviewed manuscripts related to environmental impact assessment and data-driven modeling.

Master's Thesis worker

Valmet Automotive EV power Oy

Salo, Finland

March 2022 – June 2023

- Thesis Topic: “Map out the best BI tool and SPC software: selection and rationalization process.”
- Utilized Python, Power BI, and Excel to analyze and visualize data for evaluating and selecting the optimal Business Intelligence (BI) tool and Statistical Process Control (SPC) software.
- Recommended Microsoft Power BI based on comprehensive analysis, leading to its successful adoption by Valmet as the preferred BI tool.

Laboratory and quality Operator

Valmet Automotive EV power Oy

Salo, Finland

August 2021 – November 2023

- Conducted quality checks for 48-volt battery packs, achieving a 98% quality rate.
- Analyzed production line data using Excel and visualized findings in Power BI, leading to improved decision-making.

CAREER TRANSITION AND RESEARCH ENGAGEMENT

May 2024 – Present

- Focused on enhancing skills in data science and AI through independent research and ongoing collaboration with academic peers.
- Pursuing independent research collaborations and preparing manuscripts for peer-reviewed journals.

RESEARCH OUTPUT

Published / Accepted (peer-reviewed): 10

1. Mamun, M. A., Aktar, M. N., Tasnuva, A., Islam, M. T., Mishra, M., & Islam, A. R. M. T. (2026). Decoding flood resilience in riverine Bangladesh: A concordance–discordance framework integrating AHP–TOPSIS and PROMETHEE II. *Ecohydrology & Hydrobiology*, 26(4), 100812. <https://doi.org/10.1016/j.ecohyd.2026.100812>
Contribution: Conceptualization, Methodology, Software, Visualization, Writing – original draft.
2. Mamun, M. A., Aktar, M. N., Mallick, J., Alqadhi, S., & Islam, A. R. M. T. (2026). Hybrid ensemble machine learning for groundwater level prediction in data-scarce northern Bangladesh. *Urban Water Journal*. <https://doi.org/10.1080/1573062X.2026.2649885>
Contribution: Conceptualization, Writing – review & editing, Visualization, Software, Investigation, Data curation.
3. Islam, A. R. M. T., Mamun, M. A., Tasnuva, A., Aktar, M. N., Mishra, M., Al Mamun, A., & Moin, M. J. (2026). Living on the flood line: Constructing and validating a combined multidimensional resilience index for rural riverine floodplain communities. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2026.181581>
Contribution: Writing – review & editing, Visualization, Software, Investigation, Data curation.
4. Islam, A. R. M. T., Mamun, M. A., Uddin, M. N., Sowrav, S. F. F., Alam, M. N. E., Khan, S. R., Chowdhury, M. H., & Choudhury, T. R. (2026). Explainable and physics-informed machine learning for seasonal water quality prediction in the monsoon-driven Padma River Basin, Bangladesh. *Environmental Geochemistry and Health*. <https://doi.org/10.1007/s10653-026-03031-z>
Contribution: Writing – review & editing, Visualization, Software, Investigation, Data curation.
5. Mamun, M. A., Aktar, M. N., Uddin, M. N., Chowdhury, M. H., Islam, M. S., Rahman, M. S., Uddin, M. R., Hossain, M. J., Zahid, A., Senapathi, V., & Islam, A. R. M. T. (2026). Hybrid data-driven framework for interpretable prediction of nitrate and sulfate risks in coastal aquifers. *Science of the Total Environment*, 1011, 181190. <https://doi.org/10.1016/j.scitotenv.2025.181190>
Contribution: Writing – original draft, Visualization, Software, Formal analysis, Conceptualization.

6. Islam, A. R. M. T., Varol, M., Mallick, J., **Mamun, M. A.**, Mia, M. Y., Siddique, M. A. B., Islam, M. S., & Aktar, M. N. (2026). Tracing source footprints of heavy metal(oid)s in coastal soils using traditional statistical techniques and machine learning data-driven models. *Marine Pollution Bulletin*, 222(Part 1), 118701. <https://doi.org/10.1016/j.marpolbul.2025.118701>

Contribution: Methodology, Investigation, Formal analysis, Writing – review & editing.

7. Bushra, A., **Mamun, M. A.**, & Islam, A. R. M. T. (2026). Influence of climate drivers on extreme precipitation in Bangladesh: Spatiotemporal patterns and underlying mechanisms. *International Journal of Climatology*, e70362. <https://doi.org/10.1002/joc.70362>

Contribution: Software, methodology, data analysis and review and editing.

8. Zesha, J. H., **Mamun, M. A.**, Islam, A. R. M. T., Habib, M. A., & Khan, R. (2025). First probabilistic radiological risk appraisal of Bay of Bengal beach sands: Explicit spatial hot-spot analysis. *Marine Pollution Bulletin*, 223, 118966. <https://doi.org/10.1016/j.marpolbul.2025.118966>

Contribution: Writing – review & editing, Software, Methodology, Formal analysis, Data curation.

9. Islam, A. R. M. T., **Mamun, M. A.**, Hasan, M., Aktar, M. N., Uddin, M. N., Siddique, M. A. B., Chowdhury, M. H., Islam, M. S., Bari, A. B. M. M., Idris, A. M., & Senapathi, V. (2024). Optimizing coastal groundwater quality predictions: A novel data mining framework with cross-validation, bootstrapping, and entropy analysis. *Journal of Contaminant Hydrology*, 269, 104480. <https://doi.org/10.1016/j.jconhyd.2024.104480>

Contribution: Writing – review & editing, Methodology, Formal analysis, Data curation.

10. **Mamun, M. A.**, Islam, A. R. M. T., Aktar, M. N., Uddin, M. N., Islam, M. S., Pal, S. C., Islam, A., Bari, A. B. M. M., Idris, A. M., & Senapathi, V. (2024). Predicting groundwater phosphate levels in coastal multi-aquifers: A geostatistical and data-driven approach. *Science of the Total Environment*. <https://doi.org/10.1016/j.scitotenv.2024.176024>

Contribution: Writing – original draft, Software, Methodology, Formal analysis.

REFERENCES

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